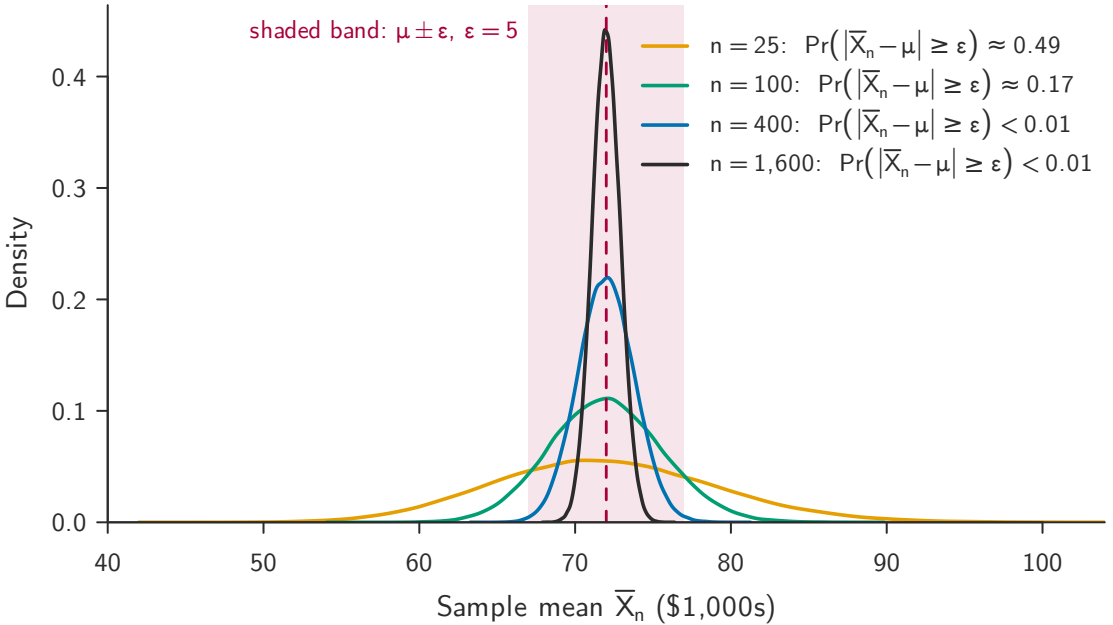
shaded band: $\mu \pm \varepsilon$, $\varepsilon = 5$
$n = 25$: $\Pr(|\bar{X}_n - \mu| \geq \varepsilon) \approx 0.49$
$n = 100$: $\Pr(|\bar{X}_n - \mu| \geq \varepsilon) \approx 0.17$
$n = 400$: $\Pr(|\bar{X}_n - \mu| \geq \varepsilon) < 0.01$
$n = 1{,}600$: $\Pr(|\bar{X}_n - \mu| \geq \varepsilon) < 0.01$
Density
0.0
0.1
0.2
0.3
0.4
40
50
60
70
80
90
100
Sample mean $\bar{X}_n$ ($1,000s)